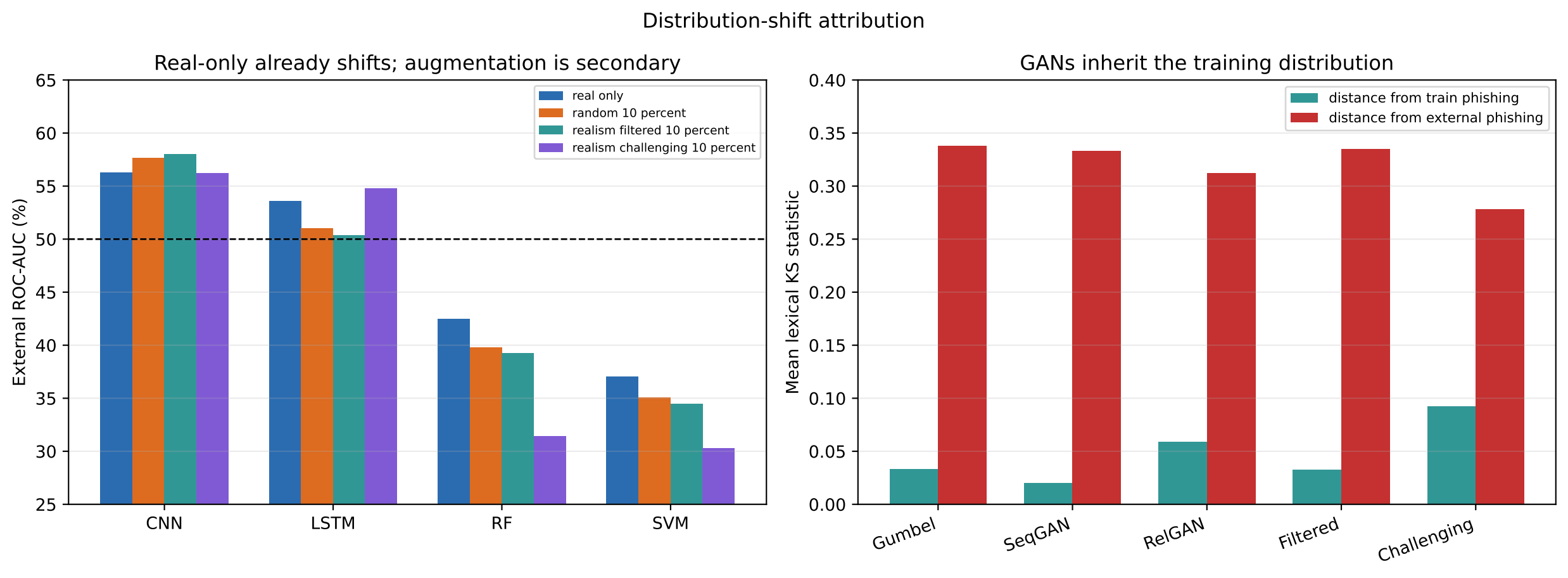
Distribution-shift attribution
Real-only already shifts; augmentation is secondary
External ROC-AUC (%)
65
60
55
50
45
40
35
30
25
real only
random 10 percent
realism filtered 10 percent
realism challenging 10 percent
CNN
LSTM
RF
SVM
GANs inherit the training distribution
Mean lexical KS statistic
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
distance from train phishing
distance from external phishing
Gumbel
SeqGAN
RelGAN
Filtered
Challenging